Concept 4: Swagger / OpenAPI

Swagger is how you write the **contract** between front-end and back-end before coding anything.

Think of it like a blueprint for a building. Before construction starts, the architect and the client agree on the plan: how many rooms, where the doors are, what goes where. Swagger is that blueprint for your API.

A Swagger document describes, for each API resource:

- **What endpoints exist** (the URIs)
- **What HTTP method** each one uses
- **What data you send** in a request (the input)
- **What data comes back** in the response (the output)
- **What errors** can happen

Here's a simplified example for your RFQ resource:

```
POST /rfq-manager/v1/rfqs
  Input:  { client_name, rfq_reference, due_date, ... }
  Output: { id, client_name, rfq_reference, status, created_at }
  Errors: 400 Bad Request, 401 Unauthorized

GET /rfq-manager/v1/rfqs/{id}
  Input:  nothing (just the id in the URL)
  Output: { id, client_name, rfq_reference, status, stages, ... }
  Errors: 404 Not Found
```

Why does this matter? Once the Swagger is agreed on, **the front-end and back-end teams can work in parallel.** The front-end knows exactly what to send and what to expect back. The back-end knows exactly what to build. Nobody waits for anybody.